

FRUHOLMEN FYR, Norway

WMO: 010550

Lat: 71.093N

Lon: 23.982E

Elev: 47

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.4	18.2	10.1	9.2	17.9	12.5	10.4	20.4	54.2	36.8	49.4	33.8	23.5	150	0.980

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	64.0	56.1	60.2	54.3	57.2	52.4	57.1	62.8	54.9	59.4	53.0	56.7	12.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.4	63.0	59.0	52.2	58.1	56.9	50.5	54.6	54.7	24.6	63.0	23.2	59.7	22.1	56.9	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 45.5	40.4	35.8	12.2	70.4	4.2	4.8	9.2	73.9	6.8	76.7	4.4	79.4	1.4	82.9		
(5)			11.8	60.8	4.2	3.6	8.8	63.4	6.3	65.5	3.9	67.5	0.8	70.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	29.8	28.8	30.6	34.0	39.4	44.5	49.9	50.2	46.9	40.2	34.8	32.3		
	DBStd	9.06	5.83	5.93	5.09	4.83	4.72	4.36	5.02	4.21	4.63	4.86	4.54	5.66		
	HDD50	4341	627	595	601	481	329	176	61	48	114	306	455	548		
	HDD65	9666	1092	1015	1066	931	792	614	468	458	542	770	905	1013		
	CDD50	152	0	0	0	0	2	12	59	55	23	1	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	4	0	0	0	0	0	0	4	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	17.6	19.7	20.4	20.1	17.0	16.9	15.6	14.1	14.0	16.5	18.2	18.5	20.8		
	PrecAvg	34.4	3.6	3.0	2.9	2.1	2.0	2.2	2.5	2.8	3.0	3.5	3.2	3.7		
(15) Precipitation	PrecMax	42.8	6.2	5.1	5.4	3.7	4.3	3.9	6.0	6.5	5.1	5.7	6.2	5.5		
	PrecMin	26.3	1.1	0.9	0.9	0.7	0.8	0.7	0.7	1.3	1.1	1.6	0.5	1.8		
	PrecStd	3.7	1.4	1.1	1.0	0.7	0.9	0.9	1.2	1.1	0.9	1.0	1.3	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	42.7	44.0	49.4	58.0	65.4	71.8	66.8	62.0	54.1	46.9	45.6		
		MCWB	40.4	39.3	39.1	42.4	48.6	55.8	60.6	57.1	54.5	49.5	44.0	43.0		
	2%	DB	41.7	41.0	41.9	44.3	52.4	58.2	65.7	62.4	58.2	50.6	44.9	43.8		
		MCWB	39.4	38.4	38.9	40.7	46.1	52.3	58.1	55.4	53.4	47.2	42.5	41.5		
	5%	DB	40.0	39.3	39.6	41.9	48.4	53.2	61.0	59.0	55.6	48.4	43.2	41.9		
		MCWB	37.8	37.0	36.9	39.4	43.9	49.2	55.0	53.9	51.4	45.8	40.8	39.1		
	10%	DB	37.9	37.0	37.8	40.5	45.4	49.4	57.0	56.2	52.9	46.9	41.2	39.8		
		MCWB	35.4	34.5	35.1	38.2	41.9	46.5	52.7	52.3	49.3	44.6	38.9	37.1		
	0.4%	WB	41.8	40.4	41.1	43.8	50.4	56.6	62.8	58.4	56.5	50.1	44.9	44.0		
		MCDB	42.7	41.8	42.4	47.3	55.8	62.7	69.4	63.5	59.7	52.7	46.3	45.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.9	38.7	39.2	41.3	46.7	52.8	58.2	56.4	54.1	47.6	43.1	42.1		
		MCDB	41.3	40.5	41.5	43.7	51.8	57.8	65.0	61.4	57.6	49.7	44.7	43.4		
	5%	WB	38.1	37.0	37.3	39.9	44.3	49.5	55.5	54.4	51.7	46.2	41.1	39.5		
		MCDB	39.9	38.9	39.5	41.7	47.7	52.9	60.7	58.4	55.1	48.0	42.7	41.3		
	10%	WB	35.9	34.8	35.4	38.4	42.6	46.9	53.0	52.7	49.8	44.8	39.1	37.3		
		MCDB	37.7	36.8	37.5	40.2	44.8	49.0	56.7	55.9	52.2	46.8	41.0	39.5		
	5% DB	MDBR	6.2	6.2	5.3	5.0	5.3	5.1	6.2	5.5	5.0	4.7	5.3	5.8		
		MCDBR	7.4	7.9	6.6	6.3	9.8	11.9	14.2	11.5	8.8	6.3	7.2	7.6		
		MCWBR	7.0	7.5	5.9	4.8	6.0	7.4	8.4	7.1	5.4	5.2	6.6	7.0		
		MCWBR	7.2	7.7	6.6	6.3	9.0	11.0	13.3	10.7	8.2	6.1	6.9	7.3		
(40) Clear-Sky Solar Irradiance	taub	N/A	0.265	0.322	0.337	0.347	0.339	0.349	0.337	0.315	N/A	N/A	N/A	N/A		
	taud	N/A	2.077	2.205	2.332	2.422	2.504	2.507	2.525	2.510	N/A	N/A	N/A	N/A		
	Ebn at Noon	0	123	206	249	263	270	262	251	220	N/A	0	0	0		
	Edh at Noon	0	11	23	28	29	28	27	23	17	N/A	0	0	0		
(44) All-Sky Solar Radiation	RadAvg	0	61	345	893	1347	1502	1331	914	474	122	0	0	0		
	RadStd	0	8	33	70	128	126	139	78	53	16	0	0	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	N/A	N/A	N/A	N/A	N/A	-214	-272	N/A	N/A
(47) Regional (10 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page